

UTSUNOMIYA, Japan

WMO: 476150

Lat: 36.5498N

Lon: 139.8680E

Elev: 140

StdP: 99.65

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.1	-3.1	-13.5	1.2	4.6	-11.9	1.4	4.5	11.0	4.8	9.4	5.7	2.3	10	0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.7	33.8	25.5	32.4	25.1	31.0	24.5	26.4	31.6	25.8	30.6	25.2	29.6	4.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.1	20.5	28.9	24.5	19.8	28.3	23.9	19.1	27.8	83.3	31.6	80.5	30.8	78.0	29.9	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.3	7.7	6.5	-6.1	35.8	1.0	1.0	-6.8	36.5	-7.4	37.1	-8.0	37.6	-8.7	38.3		
			-7.2	27.3	1.0	0.7	-7.9	27.8	-8.5	28.3	-9.0	28.7	-9.8	29.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.7	3.0	4.0	7.6	13.1	18.2	21.5	25.5	26.7	22.9	17.1	10.7	5.4
	DBStd	8.66	2.12	2.64	3.25	3.48	2.88	2.72	2.96	2.65	3.18	2.97	3.05	2.72
	HDD10.0	662	217	168	92	12	0	0	0	0	0	28	144	
	HDD18.3	2104	475	400	334	160	38	5	0	3	60	229	400	
	CDD10.0	2379	0	1	16	104	253	346	482	518	387	219	49	3
	CDD18.3	781	0	0	0	2	33	101	224	259	141	21	0	0
	CDH23.3	5988	0	0	1	22	202	546	1864	2401	884	69	0	0
	CDH26.7	2059	0	0	0	2	33	126	706	933	251	9	0	0
Wind	WSAvg	2.9	2.9	3.0	3.3	3.3	3.1	2.8	2.7	2.8	3.0	2.8	2.6	2.7
Precipitation	PrecAvg	1631	44	47	85	127	154	177	222	217	255	168	88	48
	PrecMax	2078	129	122	161	232	346	482	366	468	432	508	230	163
	PrecMin	1126	5	6	18	28	71	57	68	59	28	12	1	5
	PrecStd	214	31	29	36	61	63	82	80	109	107	106	53	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.4	16.8	21.3	25.7	29.6	31.9	35.1	35.4	33.0	27.8	21.0	16.1
		MCWB	7.6	10.3	11.8	15.1	18.7	23.1	25.7	26.3	24.8	20.7	14.6	10.6
	2%	DB	10.8	13.1	18.2	23.3	27.3	29.6	33.7	34.0	31.1	25.1	19.1	13.9
		MCWB	5.5	7.2	10.6	14.4	17.6	21.7	25.5	25.8	23.9	19.1	13.6	9.1
	5%	DB	9.3	11.2	16.0	21.4	25.5	28.0	32.2	32.8	29.5	23.4	17.7	12.3
		MCWB	4.1	5.5	9.5	13.8	17.1	20.8	25.0	25.5	23.2	17.9	12.8	7.8
	10%	DB	8.0	9.5	13.9	19.5	23.9	26.5	30.7	31.5	27.8	22.0	16.2	10.8
		MCWB	3.3	4.3	8.4	13.1	16.7	20.3	24.5	25.1	22.5	16.8	11.9	6.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	11.3	14.0	19.1	21.5	24.7	27.1	27.5	25.9	22.0	16.6	12.7
		MCDB	11.1	15.2	17.4	21.2	25.6	29.8	32.9	33.1	30.4	26.2	18.8	14.4
	2%	WB	6.6	8.2	12.3	16.7	19.8	23.5	26.2	26.7	25.0	20.5	15.2	10.3
		MCDB	9.6	11.7	16.2	21.0	24.1	27.4	31.6	32.0	28.8	23.5	17.5	12.7
	5%	WB	5.0	6.6	10.8	15.2	18.8	22.5	25.6	26.1	24.3	19.3	14.1	8.5
		MCDB	8.6	9.9	14.7	19.1	23.0	26.2	30.6	31.0	27.8	21.9	16.3	11.3
	10%	WB	3.7	5.0	9.1	14.0	17.8	21.5	25.0	25.5	23.6	18.1	12.9	7.2
		MCDB	7.2	8.7	13.1	17.6	21.9	25.0	29.6	30.0	26.8	21.0	15.2	10.2
Mean Daily Temperature Range	5% DB	MDBR	10.0	10.1	10.3	10.5	9.3	7.7	7.5	7.7	7.5	8.1	9.3	9.8
		MCDBR	11.6	12.4	13.3	13.2	12.2	10.3	9.9	9.5	9.1	9.5	10.5	11.1
		MCWBR	7.1	7.7	7.7	6.7	5.5	4.4	3.8	3.5	3.7	4.8	6.3	6.8
	5% WB	MCDBR	10.4	10.9	11.3	10.4	9.7	8.3	9.1	8.9	8.1	7.8	8.4	9.4
		MCWBR	7.1	7.5	7.8	6.5	5.3	4.0	3.6	3.5	3.7	4.6	5.9	6.6
Clear-Sky Solar Irradiance	taub	0.362	0.409	0.482	0.537	0.564	0.604	0.601	0.595	0.513	0.462	0.403	0.365	
	taud	2.237	2.093	1.925	1.851	1.838	1.807	1.870	1.876	2.051	2.114	2.207	2.266	
	Ebn at Noon	807	807	778	759	746	714	713	706	744	743	751	776	
	Edh at Noon	105	137	178	202	208	214	200	195	156	133	108	97	
All-Sky Solar Radiation	RadAvg	2.61	3.26	4.09	4.76	5.14	4.65	4.67	4.62	3.72	3.05	2.49	2.28	
	RadStd	0.14	0.24	0.23	0.41	0.51	0.36	0.73	0.51	0.30	0.29	0.21	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page